

SFANTU GHEORGHE, Romania

WMO: 152380

Lat: 45.8717N

Lon: 25.8022E

Elev: 1720

StdP: 13.81

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	6.8	-1.9	5.3	1.9	3.1	6.9	8.2	20.6	31.8	16.5	27.6	0.8	30	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	87.7	67.1	84.9	66.2	82.1	65.5	69.7	82.8	68.3	80.7	66.9	78.9	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	100.9	73.9	64.2	95.9	72.4	62.9	91.6	70.8	34.8	82.8	33.6	80.5	32.4	79.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	12.0	9.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	25.5	31.3	38.2	47.4	57.6	64.6	66.5	67.0	58.2	48.2	39.6	30.4		
	DBStd	15.93	8.86	9.11	7.75	8.34	5.60	5.25	4.73	4.92	7.22	6.44	7.34	7.98		
	HDD50	2865	759	524	369	144	11	1	0	0	15	112	321	609		
	HDD65	6487	1224	944	830	527	234	69	41	40	220	521	763	1074		
	CDD50	2120	0	0	4	68	247	439	512	526	261	55	8	0		
	CDD65	268	0	0	0	0	5	58	88	101	16	0	0	0		
	CDH74	3806	0	0	0	43	188	780	1062	1354	366	13	0	0		
	CDH80	1120	0	0	0	1	23	204	331	471	90	0	0	0		
	WSAvg	2.9	2.7	2.7	3.7	3.5	3.1	2.9	3.0	2.7	2.8	2.5	2.3	2.5		
	PrecAvg	23.0	0.9	0.9	1.1	2.0	3.0	3.5	3.3	2.5	2.0	1.7	1.1	1.1		
(15) Precipitation	PrecMax	33.4	2.1	1.4	2.0	3.6	6.8	6.0	6.0	4.7	4.0	3.8	2.3	2.6		
	PrecMin	16.7	0.2	0.2	0.4	0.7	1.3	0.9	1.4	0.9	0.3	0.1	0.0	0.2		
	PrecStd	3.9	0.4	0.3	0.4	0.9	1.3	1.5	1.4	1.2	0.9	0.9	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	60.7	68.4	78.1	83.0	88.9	90.4	91.8	86.7	75.7	65.4	51.3		
		MCWB	44.1	50.3	51.3	57.2	61.4	69.4	68.4	68.5	64.8	59.2	54.0	46.2		
	2%	DB	42.9	53.1	61.6	74.2	78.9	85.0	87.1	88.0	82.0	71.3	60.8	46.4		
		MCWB	39.5	45.5	48.3	55.9	60.8	67.4	67.3	66.6	63.1	57.2	52.1	42.9		
	5%	DB	39.1	48.3	56.1	69.6	75.4	81.8	83.6	85.0	78.1	66.9	56.0	42.7		
		MCWB	36.5	42.5	45.6	54.3	60.2	66.5	66.7	65.8	62.0	55.6	49.7	40.0		
	10%	DB	36.4	43.0	51.4	64.1	71.8	78.2	80.1	81.7	74.1	61.3	51.8	39.4		
		MCWB	34.3	38.9	43.1	52.0	58.2	64.5	65.6	65.4	60.9	53.3	46.6	37.5		
	0.4%	WB	44.0	50.9	52.6	59.3	64.9	72.1	70.9	71.2	67.0	60.9	55.5	47.3		
		MCDB	48.0	60.3	67.6	75.2	76.8	85.2	83.3	86.7	82.4	71.1	63.7	50.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.9	46.0	49.4	57.2	62.8	69.2	69.4	69.4	64.8	58.5	53.0	43.0		
		MCDB	42.6	53.1	60.7	70.8	74.8	81.9	81.8	83.1	78.1	69.2	59.6	46.4		
	5%	WB	37.1	42.9	46.7	55.2	61.1	67.5	68.1	67.8	63.1	56.5	50.7	40.2		
		MCDB	39.1	48.0	55.8	68.3	72.9	79.5	79.7	80.9	75.2	65.0	56.7	42.7		
	10%	WB	34.8	39.6	44.2	52.9	59.5	65.7	66.7	66.5	61.6	54.1	47.5	37.8		
		MCDB	36.3	42.6	50.5	62.8	70.0	76.2	78.0	78.5	72.7	60.8	52.0	39.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	18.1	19.8	22.8	23.4	23.8	23.8	26.8	24.1	22.7	17.8	12.1		
		MCDBR	15.9	24.5	30.9	34.7	31.4	29.9	30.9	33.0	32.9	34.7	27.0	16.3		
		MCWBR	12.5	16.6	18.0	17.2	14.7	13.3	12.6	12.9	14.7	20.4	17.7	13.0		
	5% WB	MCDBR	15.2	24.4	28.9	31.0	28.1	27.6	26.6	29.1	29.7	31.1	25.8	15.6		
		MCWBR	12.0	16.9	17.1	15.9	13.6	13.1	11.6	12.3	13.4	18.7	16.9	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.332	0.376	0.435	0.414	0.412	0.430	0.442	0.397	0.359	0.332	0.304		
	taud		2.335	2.266	2.223	2.100	2.232	2.272	2.228	2.189	2.285	2.381	2.405	2.372		
	Ebn at Noon		251	264	268	260	271	272	265	256	258	252	239	238		
	Edh at Noon		24	32	38	47	43	42	43	43	36	28	23	21		
(44) All-Sky Solar Radiation	RadAvg		425	686	1036	1376	1703	1830	1848	1681	1217	824	469	342		
	RadStd		49	82	111	143	157	158	154	138	122	101	49	41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)		+1.12	N/A	N/A	+1.24	N/A	N/A	-248	-329	+139	+43

Nomenclature: See separate page